

Design and Simulation of a 2.4 GHz Rectangular Microstrip Patch Antenna

An FDTD Electromagnetic Simulation Study using openEMS

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Abstract

This report presents the design, electromagnetic simulation, and performance analysis of a rectangular microstrip patch antenna operating at 2.4 GHz, intended for ISM-band applications such as Wi-Fi, Bluetooth, and IoT devices. The antenna uses an inset microstrip feed on an FR4 substrate and was simulated using the open-source FDTD electromagnetic solver openEMS, driven via its Python (CSXCAD) interface. The final design achieves a simulated resonance at 2.405 GHz, a minimum S11 of -13.71 dB, VSWR of 1.52, simulated directivity of 6.56 dBi, and gain of 6.36 dBi, with a -10 dB impedance bandwidth of 28.86 MHz and a VSWR ≤ 2 bandwidth of 35.37 MHz. The reported 95.51% radiation efficiency reflects an idealized simulation model (perfect electric conductor metal, lossless dielectric) and is discussed accordingly. Radiation characteristics are presented as E-plane and H-plane polar patterns.

1. Introduction

Microstrip patch antennas are widely used in compact wireless systems due to their low profile, light weight, ease of fabrication using standard printed-circuit techniques, and compatibility with planar RF front ends. This project designs and simulates a rectangular microstrip patch antenna operating in the 2.4 GHz Industrial, Scientific and Medical (ISM) band, which is used by Wi-Fi (802.11b/g/n), Bluetooth, Zigbee, and a wide range of IoT devices.

The objective of this project is to apply antenna theory and full-wave electromagnetic simulation to design a patch antenna, verify its resonant frequency and impedance matching, and characterize its radiation behavior through simulation, using the open-source FDTD solver openEMS. This is an academic simulation study; the antenna was not fabricated or measured, and reported results are simulation outputs under the stated modeling assumptions.

2. Antenna Theory

2.1 Operating Principle

A rectangular microstrip patch antenna consists of a thin conducting patch on one side of a dielectric substrate, with a ground plane on the other side. The patch is commonly modeled as a resonant cavity bounded by radiating slots along two edges, and it operates approximately as a half-wavelength resonator at the design frequency, with the effective electrical length modified by the substrate's dielectric properties and by fringing fields at the patch edges.

2.2 Effective Dielectric Constant and Fringing

Because the electromagnetic fields of a microstrip structure exist partly in the dielectric substrate and partly in the air above it, an effective dielectric constant (ϵ_{eff}), intermediate between the substrate's relative permittivity and that of air, is used to determine the patch's electrical length. The physical patch length is shorter than the free-space half-wavelength because the fringing fields at the radiating edges electrically extend the patch beyond its physical boundary; this extension (ΔL) is subtracted when moving from the effective length back to the physical length.

2.3 Feeding and Impedance Matching

A patch antenna's input impedance varies with position along its length, being highest at the radiating edges and passing through 50 ohms at some inset distance from the edge. An inset microstrip feed exploits this by recessing

the feed line into the patch by a chosen depth, allowing the feed to connect at a point where the patch's edge impedance matches the standard 50 ohm system impedance, without requiring a separate matching network.

3. Antenna Design and Parameters

The final antenna geometry, substrate, and feed parameters used in the openEMS simulation are summarized below. The patch was designed for 2.4 GHz resonance on FR4 substrate, with an inset microstrip feed for direct 50 ohm matching.

Parameter	Value
Design frequency	2.4 GHz
Substrate material	FR4
Substrate relative permittivity (epsilon_r)	4.3
Substrate dimensions (W x L x h)	58 mm x 65 mm x 1.6 mm
Patch dimensions (W x L)	38.37 mm x 28.95 mm
Feed type	Inset microstrip line
Feed dimensions (W x L)	2.0 mm x 10.0 mm
Inset depth	8.0 mm
Inset gap (each side)	1.0 mm
Conductor model	Zero-thickness perfect electric conductor (PEC)
Port	50 ohm lumped port

The ground plane, patch, and feed conductors were modeled as zero-thickness PEC surfaces rather than volumetric copper. This is a standard and physically justified simplification at 2.4 GHz: the skin depth of copper at this frequency is approximately 1.3 micrometres, far thinner than a typical 35 micrometre copper layer, so current is already confined to a thin surface layer and a zero-thickness PEC boundary closely approximates the physical conductor while substantially improving FDTD mesh efficiency and simulation runtime.

4. Simulation Setup

The antenna geometry, materials, and feed were defined in Python using the CSXCAD geometry library, and the electromagnetic simulation was performed using openEMS, an open-source Finite-Difference Time-Domain (FDTD) solver. The simulation setup consisted of the following elements:

- A non-uniform FDTD mesh with explicit lines at all geometry edges (patch, feed, inset notch, substrate boundaries), refined near metal edges using automatic edge-to-grid alignment, and smoothed with a maximum cell size of 2 mm.
- Perfectly Matched Layer (PML) absorbing boundary conditions, 8 cells thick, on all six faces of the simulation domain, to absorb outgoing radiation without significant reflection.
- A 50 ohm lumped port at the feed line, exciting the structure with a Gaussian pulse centered at 2.4 GHz with 1.0 GHz bandwidth.

- A near-field-to-far-field (NF2FF) recording box enclosing the antenna, created prior to running the solver, used in post-processing to compute far-field radiation patterns, directivity, and gain.
- A maximum of 30,000 timesteps with an energy-decay end criterion of -40 dB (1e-4), used to terminate the simulation once the fields had sufficiently decayed.

Post-processing was performed in a separate analysis script that reads the recorded port and NF2FF field data to compute S11, VSWR, directivity, gain, radiation efficiency, and the E-plane and H-plane radiation patterns.

5. Results

5.1 Impedance Matching: S11

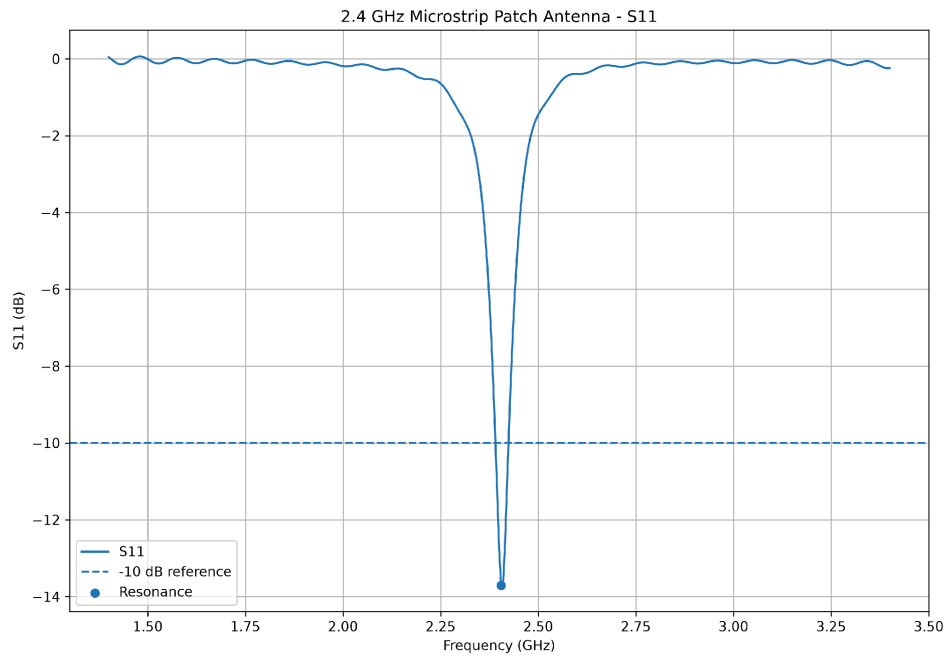


Figure 1: Simulated S11 (return loss) versus frequency, 1.4-3.4 GHz.

The simulated S11 shows a clear resonance dip at 2.405 GHz, very close to the 2.4 GHz design target (0.21% frequency error). The minimum S11 is -13.71 dB, comfortably below the commonly used -10 dB matching criterion, indicating that approximately 96% of the incident power at resonance is delivered to the antenna rather than reflected back to the source.

5.2 Voltage Standing Wave Ratio (VSWR)

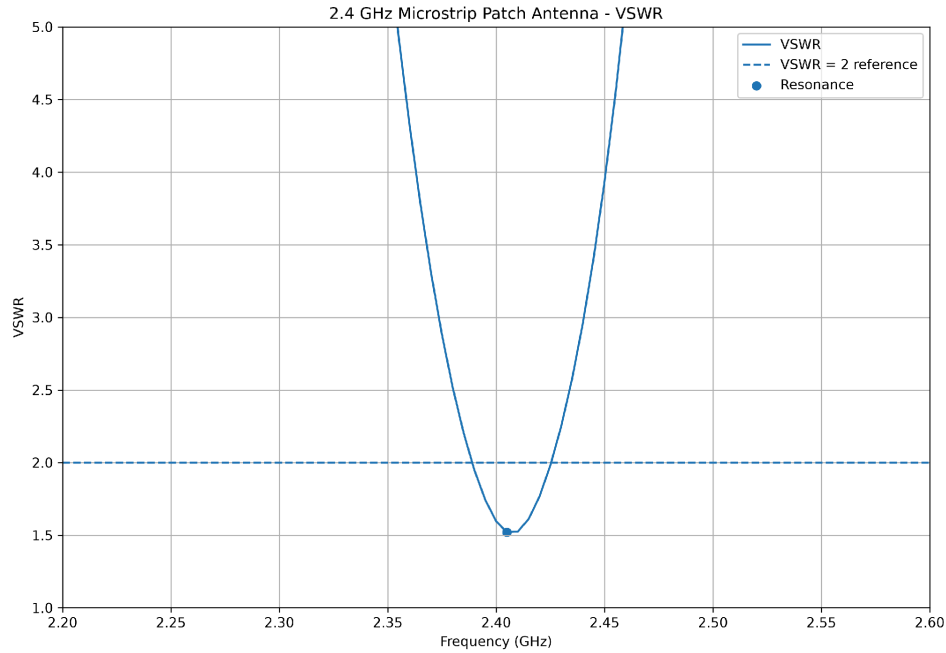


Figure 2: Simulated VSWR versus frequency, shown over the 2.2-2.6 GHz band of interest.

The VSWR reaches a minimum of 1.52 at 2.405 GHz, well below the $\text{VSWR} \leq 2$ threshold commonly used as a practical matching criterion. This is consistent with the S11 result, since both are derived from the same reflection coefficient.

5.3 Impedance Bandwidth

Bandwidth was computed directly from the simulated S11 and VSWR data by locating the frequencies at which each curve crosses its respective threshold on either side of resonance.

Criterion	Lower / Upper Frequency	Bandwidth	Fractional Bandwidth
S11 ≤ -10 dB	approx. 2.376 - 2.405 GHz range (from data)	28.86 MHz	1.20 %
VSWR ≤ 2	approx. 2.389 - 2.425 GHz range (from data)	35.37 MHz	1.47 %

The narrow bandwidth (on the order of 1-1.5%) is typical for a thin ($h = 1.6$ mm) FR4 patch antenna and is further narrowed by the lossless dielectric model used in this simulation; a lossy FR4 substrate (nonzero loss tangent) would be expected to broaden the resonance somewhat at the cost of reduced peak efficiency.

5.4 Radiation Pattern: E-plane and H-plane

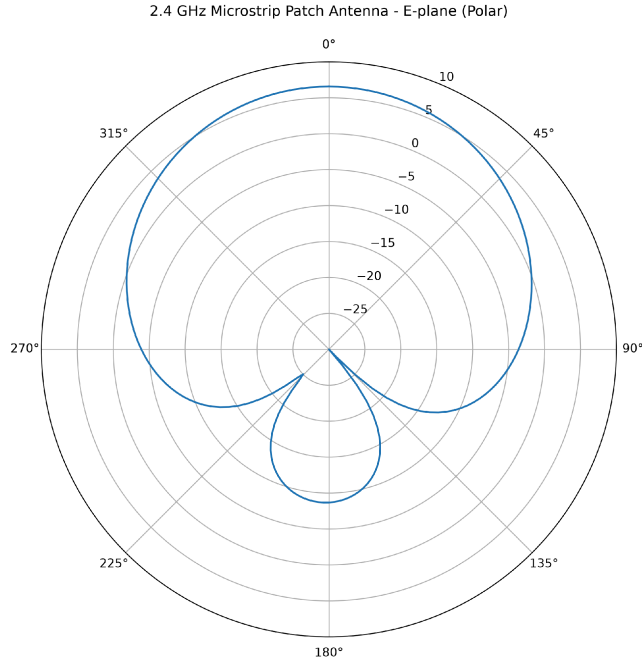


Figure 3: E-plane (YZ, $\phi = 90$ deg) radiation pattern, polar plot, directivity in dBi (-30 dB floor).

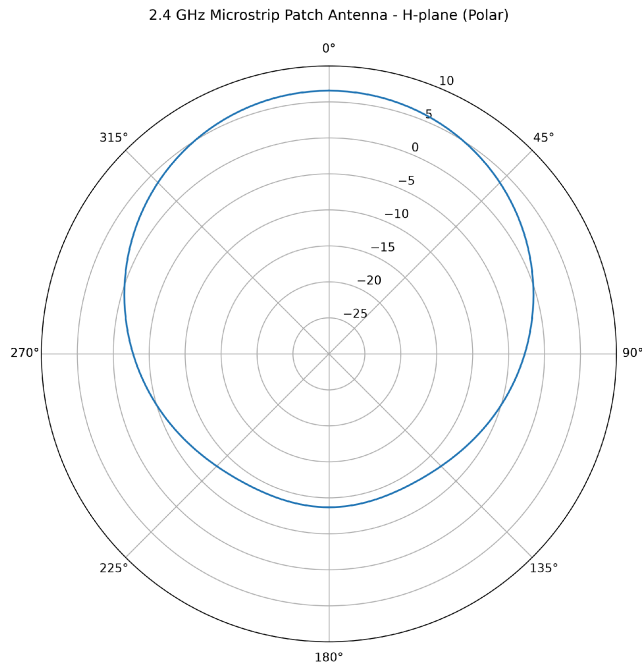


Figure 4: H-plane (XZ, $\phi = 0$ deg) radiation pattern, polar plot, directivity in dBi (-30 dB floor).

Both principal-plane patterns show a dominant main lobe pointing broadside ($\theta = 0$ degrees, +Z direction), consistent with expected patch antenna behavior. The E-plane pattern shows a peak directivity of 6.56 dBi at broadside, deep nulls near 138 degrees and 222 degrees caused by ground-plane edge diffraction, and a smaller back lobe near $\theta = 180$ degrees at approximately -8.7 dBi, giving a front-to-back ratio of approximately 15 dB. The H-plane pattern is smoother, without the sharp nulls seen in the E-plane, which is expected since the two planes correspond to different current distributions on the patch.

The 3 dB beamwidth, computed directly from the simulated pattern data, is 90.48 degrees in the E-plane and 91.37 degrees in the H-plane.

5.5 Directivity, Gain, and Radiation Efficiency

Parameter	Value
Radiated power	2.5275e-26 W
Accepted input power	2.6463e-26 W
Directivity	6.56 dBi
Radiation efficiency	95.51 %
Gain	6.36 dBi

Gain, directivity, and efficiency are self-consistent: $\text{Gain (dBi)} = \text{Directivity (dBi)} + 10 \cdot \log_{10}(\text{efficiency})$, giving $6.557 + 10 \cdot \log_{10}(0.9551) = 6.36$ dBi, matching the simulated value. The very small absolute power values (order $1\text{e-}26$ W) are an artifact of the solver's internal excitation normalization and do not affect the efficiency ratio, which depends only on the ratio of radiated to accepted power.

6. Discussion and Limitations

The simulated 95.51% radiation efficiency should be interpreted with care. The antenna model uses perfect electric conductor (PEC) metal, which has zero conductor loss, and an FR4 substrate defined only by its relative permittivity ($\epsilon_r = 4.3$) with no dielectric loss tangent specified, meaning the substrate is modeled as lossless. With both loss mechanisms absent, the only physical energy sink in the simulation is radiation itself, so the efficiency figure is better understood as a measure of how completely the FDTD simulation's stored energy had radiated away by the time the solver's end criterion (-40 dB) was reached, rather than as a prediction of a fabricated antenna's real-world efficiency. Real FR4 has a non-negligible loss tangent (typically on the order of 0.02 at these frequencies), which would reduce efficiency and broaden the impedance bandwidth in an actual fabricated and measured device.

For the same reason, the reported gain and directivity values should be understood as outputs of this idealized simulation model rather than as guaranteed hardware performance figures. The S11, VSWR, and radiation pattern shape (main lobe direction, beamwidth, null positions, front-to-back ratio), by contrast, are geometric and resonance properties primarily governed by the patch and feed dimensions and are expected to transfer more directly to a fabricated prototype.

A three-dimensional radiation pattern surface was initially generated but was found to visually compress the true 15 dB front-to-back ratio into a shape that appeared nearly spherical when plotted on a decibel radial scale, which could be visually misleading despite being numerically correct. The E-plane and H-plane polar patterns presented in Section 5.4 were used instead, as they convey the same directivity data in a standard, quantitatively readable format.

7. Conclusion

A rectangular microstrip patch antenna was designed and simulated for 2.4 GHz operation using an inset-fed configuration on FR4 substrate, with electromagnetic performance evaluated in openEMS using the FDTD method. The final design resonates at 2.405 GHz (0.21% error from the 2.4 GHz target), achieves a minimum S11 of -13.71 dB and VSWR of 1.52, both comfortably meeting standard matching criteria ($S_{11} < -10$ dB, $VSWR < 2$), with a -10 dB bandwidth of 28.86 MHz and $VSWR \leq 2$ bandwidth of 35.37 MHz. Simulated directivity and gain are 6.56 dBi and 6.36 dBi respectively, with a dominant broadside main lobe and 3 dB beamwidths of approximately 90 degrees in both principal planes. The reported 95.51% radiation efficiency reflects an idealized lossless simulation model

and should be read as a convergence and efficiency indicator within that model rather than a hardware prediction. Overall, the simulated design meets its impedance-matching and radiation-pattern objectives for 2.4 GHz ISM-band use.

References

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